

Data Dictionary & Team Contribution

DBAD 4000 | Class Activity: Designing a Logical Database

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Gym logical database: six entities and 26 attributes. The dictionary includes booking cancellation and completed payments only. Business rules, design and rationale are provided in README.md.

Conventions

- **PK:** Primary key. **FK:** Foreign key. **UQ:** Unique alternate key.
- All fields are required except Person.phone.
- Enrollment.is_cancelled defaults to FALSE. No other field defaults are specified by this logical model.
- Data types are proposed logical SQL types. Referencing and referenced keys both use INTEGER.
- All payment amounts are in CAD. TIMESTAMP values represent gym local date and time in America/Edmonton.
- PK and FK designations on both Enrollment IDs mean one composite primary key, not two independently unique keys.

1 Person

One individual known to the gym, with shared identity and contact details.

Attribute	Data type	Key	Nullable	Definition and validation
person_id	INTEGER	PK	No	Unique identifier; positive integer; required.
first_name	VARCHAR(50)	-	No	Given name; required and not blank.
last_name	VARCHAR(50)	-	No	Family name; required and not blank.
email	VARCHAR(100)	UQ	No	Contact email; required, valid format and unique after trim and lowercase normalization; maximum 100 characters.
phone	VARCHAR(20)	-	Yes	Optional contact number; if supplied, must use a valid phone-number format for its region; maximum 20 characters. Stored as text to preserve + and leading zeros.

2 Member

Membership role held by a Person; person_id is inherited from the supertype.

Attribute	Data type	Key	Nullable	Definition and validation
person_id	INTEGER	PK, FK	No	References Person.person_id; exactly one parent Person; required.
join_date	DATE	-	No	Date this membership began; required.
membership_status	VARCHAR(20)	-	No	Current membership state; required; allowed values are exactly active and inactive. Does not represent booking or payment status.

3 Trainer

Trainer role held by a Person; only one primary certification is in scope.

Attribute	Data type	Key	Nullable	Definition and validation
person_id	INTEGER	PK, FK	No	References Person.person_id; exactly one parent Person; required.
primary_certification	VARCHAR(100)	-	No	Name of the primary certification; one value, required and not blank.
hire_date	DATE	-	No	Date the trainer began working for the gym; required.

4 ClassSession

One scheduled class occurrence; repeated classes have separate session records.

Attribute	Data type	Key	Nullable	Definition and validation
class_id	INTEGER	PK	No	Unique session identifier; positive integer; required.
trainer_id	INTEGER	FK	No	References Trainer.person_id; the one assigned trainer; required.
class_name	VARCHAR(100)	-	No	Label for this session, such as Yoga; required, not blank and not unique.
starts_at	TIMESTAMP	-	No	Session start date and time; required; gym local time in America/Edmonton.

duration_minutes	INTEGER	-	No	Scheduled length in whole minutes; required; greater than zero.
capacity	INTEGER	-	No	Maximum number of non-cancelled bookings for the session; required positive integer. Count only Enrollment rows with is_cancelled = FALSE.

5 Enrollment

Association between a Member and a ClassSession; the two IDs form one composite key. One row represents the current booking state for that member-session pair, including when the booking has been cancelled.

Composite primary key: (member_id, class_id). Neither column must be unique by itself; only the pair is unique.

Attribute	Data type	Key	Nullable	Definition and validation
member_id	INTEGER	PK part, FK	No	References Member.person_id; required; first part of the composite PK.
class_id	INTEGER	PK part, FK	No	References ClassSession.class_id; required; second part of the composite PK.
enrolled_at	TIMESTAMP	-	No	Date and time the Enrollment row was originally created; required; gym local time. Retained when the booking is cancelled or restored.
is_cancelled	BOOLEAN	-	No	Whether this booking is currently cancelled; required; default FALSE; allowed values TRUE or FALSE only. FALSE counts towards session capacity. TRUE keeps the record but releases its place. Independent of Member.membership_status.

Cancellation and re-enrollment: Set is_cancelled to TRUE to cancel without deleting the row. To re-enroll in the same session, set it back to FALSE only after checking that the member is active and the session has a place available. Reuse the existing composite-key row and preserve enrolled_at. This field does not retain a complete history of changes or distinguish attendance, absence and completion.

6 Payment

One completed member payment. Pending, failed and cancelled payment attempts are not stored; invoices, refunds and card credentials are also outside scope. All stored Payment rows already represent completed transactions, so payment_status and a separate attempt-creation timestamp are not part of this model.

Attribute	Data type	Key	Nullable	Definition and validation
payment_id	INTEGER	PK	No	Unique payment identifier; positive integer; required.

member_id	INTEGER	FK	No	References Member.person_id; the paying member; required.
paid_at	TIMESTAMP	-	No	Date and time payment was completed; required; gym local time.
amount	DECIMAL(10,2)	-	No	Amount received in CAD; required; greater than zero.
payment_method	VARCHAR(30)	-	No	Method used for the completed payment; required; allowed values are exactly cash, debit and credit.

Foreign key reference map

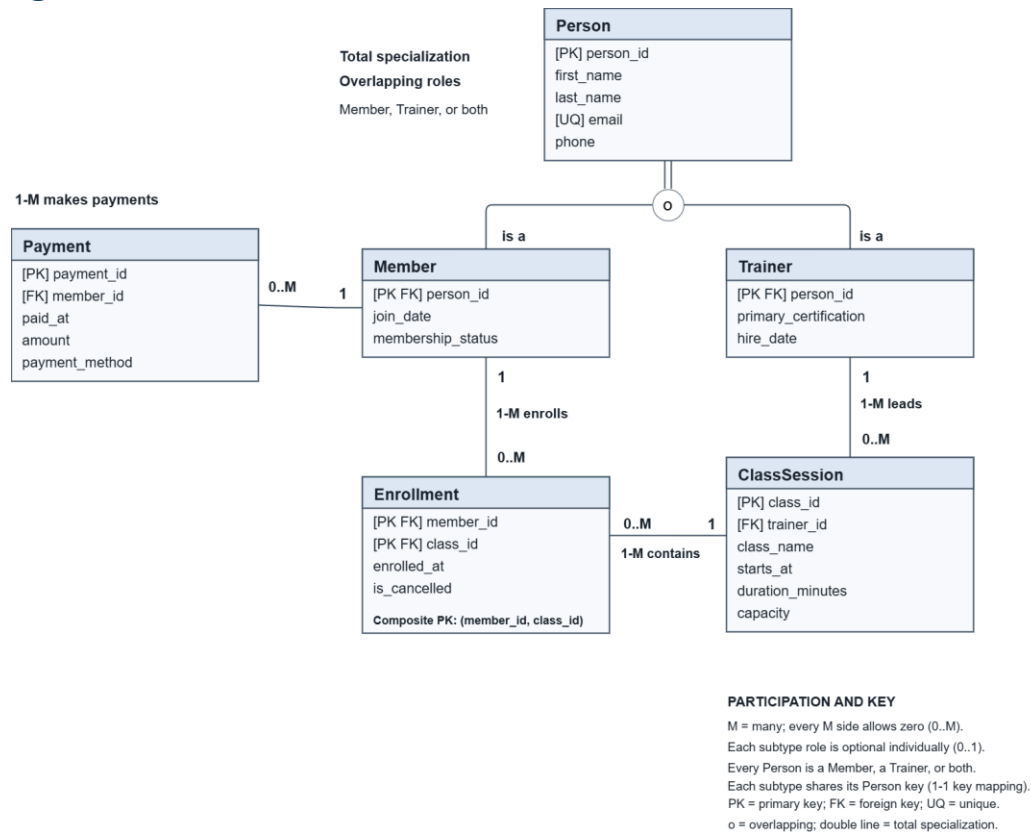
Foreign key	Referenced key	Required
Member.person_id	Person.person_id	Yes
Trainer.person_id	Person.person_id	Yes
ClassSession.trainer_id	Trainer.person_id	Yes
Enrollment.member_id	Member.person_id	Yes
Enrollment.class_id	ClassSession.class_id	Yes
Payment.member_id	Member.person_id	Yes

Entity and cross-record constraints

Constraint	Definition
Person role coverage	Every Person must have a Member row, a Trainer row, or both. Each subtype can contain at most one row for that person_id.
Canonical email uniqueness	Trim and lowercase email values before checking uniqueness; also validate email format.
Enrollment uniqueness	No two Enrollment rows may have the same member_id and class_id pair, even if one is cancelled. Restore the existing row when re-enrolling in the same session.
Enrollment eligibility	Only an active Member may create a new enrollment or restore a cancelled enrollment; later inactivation does not invalidate historical rows.
Cancellation state	is_cancelled is required, defaults to FALSE and permits only TRUE or FALSE. Cancellation sets TRUE and preserves the row. A membership-status change does not automatically change this field.

Session capacity	For each class_id, COUNT of Enrollment rows WHERE is_cancelled = FALSE must be less than or equal to ClassSession.capacity. Check new bookings, restored bookings and capacity reductions transactionally. Cancelled records do not consume places.
Original enrollment time	enrolled_at records initial row creation and is retained during cancellation and restoration; a full booking-event history is outside scope.
Status domain	membership_status must be active or inactive.
Payment method domain	payment_method must be cash, debit or credit.
Completed payments only	Create a Payment row only for a completed transaction; paid_at must be present and amount must be the positive amount received in CAD. Do not store pending, failed or cancelled attempts.
Positive values	IDs, duration_minutes, capacity and amount must be greater than zero.
Preserve history	Keep existing enrollment and completed payment records when a membership becomes inactive. Cancelling a booking does not change membership_status or delete completed payments; avoid deletion that breaks referenced records.

EERD Diagram



Team Contribution

The following table preserves the team members and contributions recorded in the supplied activity document. Wording has been lightly edited for readability; the assigned work is unchanged.

Team member	Role / contribution
Troy Franks	Set up GitHub repository; business rules.
Seth Garciano	EERD and proofreading.
Tristin Bates	First draft Entity listing, data interaction rules and data dictionary
Yisong Wang	Business rules; EERD based on the business rules; relationships and participation; normalization and design rationale and redoing the document.

AI assistance

OpenAI Codex supplemented and refined the business rules and data dictionary, assisted with document formatting and normalization explanations, and refined the EERD. The contribution records above were supplied by the team.

Companion files

README.md contains the business rules, design decisions, relationships and 3NF rationale. gym-eerd.drawio contains the editable model. The dictionary includes Enrollment.is_cancelled, and Payment remains limited to completed transactions.